TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	83.8 ± 1.72
Correlation Stability	86.7 ± 1.19
Distribution Consistency	86.0 ± 2.76
Entropy Preservation	88.8 ± 0.60
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.3 ± 0.64
Feature Accuracy	90.5 ± 1.75
Novelty Score	0.8 ± 0.01

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	54.876	+0.437	0.8%	OK
sex	0.680	0.694	+0.014	2.1%	OK
cp	3.158	3.264	+0.106	3.4%	OK
trestbps	131.690	129.124	-2.566	1.9%	OK
chol	246.693	245.702	-0.991	0.4%	OK
fbs	0.149	0.112	-0.037	24.9%	Warn
restecg	0.990	1.004	+0.014	1.4%	OK
thalach	149.607	151.926	+2.318	1.5%	OK
exang	0.327	0.302	-0.025	7.7%	OK
oldpeak	1.040	0.743	-0.297	28.5%	Drift
slope	1.601	1.624	+0.023	1.5%	OK
ca	0.672	0.562	-0.110	16.4%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	83.8 ± 1.72
Correlation Stability (*2)	86.7 ± 1.19
Distribution Consistency (*3)	86.0 ± 2.76
Entropy Preservation (*4)	88.8 ± 0.60
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.3 ± 0.64
Feature Accuracy (*10)	90.5 ± 1.75
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5770491803278688	0.5901639344262295	-0.008196721311475419	
F1	0.5650748554025102	0.48875732788698834	0.516510468970284	-0.04856438643222617	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.539344262295082	0.5459016393442623	-0.008196721311475396	
F1	0.4442202071436568 4	0.3893032298176541 4	0.4099502430668558	-0.034269964076801	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6065573770491803	0.5327868852459016	0.5918032786885247	-0.014754098360655731	
F1	0.5581883691194175	0.459305827472957	0.52938570890364	-0.028802660215777414	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5327868852459016	0.5081967213114755	0.5163934426229508	-0.016393442622950817	
F1	0.5321068042920665	0.491928244515156	0.5121196392383264	-0.019987165053739996	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5344262295081967	0.5442622950819671	0.5721311475409837	0.03770491803278691	*
F1	0.522698657145684	0.4984728600207252	0.5416005223681554	0.01890186522247137	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.5655737704918034	0.5901639344262295	-0.01803278688524592	
F1	0.6070236204175147	0.5081576136169045	0.5580072864324185	-0.04901633398509664	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.5311475409836065	0.5426229508196722	-0.01475409836065716	
F1	0.568660979990255	0.5174148381191435	0.5330394327837273	-0.03562154720652764	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.5688524590163934	0.5819672131147541	-0.01639344262295083	
F1	0.5545405481459582	0.4629375138300243	0.5155841559054988	-0.03895639224045937	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5344262295081967	0.5573770491803278	0.5786885245901640	0.04426229508196722	
F1	0.5119295254727356	0.4994679908843262	0.5503413852626382	0.03841185978990252	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5475409836065573	0.5475409836065573	0	
F1	0.3899270600110827	0.3899270600110827	0.3899270600110827	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	68.0	1.0	3.0	128.0	243.0	0.0	2.0	152.0	0.0	2.0
2	58.0	1.0	3.0	120.0	216.0	0.0	0.0	165.0	0.0	1.5
3	69.0	1.0	4.0	120.0	268.0	1.0	0.0	125.0	1.0	0.0
4	52.0	1.0	4.0	140.0	204.0	0.0	0.0	120.0	0.0	0.0
5	41.0	1.0	4.0	140.0	271.0	0.0	2.0	162.0	0.0	0.0
6	34.0	0.0	3.0	130.0	263.0	0.0	2.0	154.0	0.0	0.8
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	1.0	3.0	110.0	240.0	0.0	0.0	143.0	0.0	0.0
2	54.0	1.0	3.0	140.0	234.0	0.0	0.0	153.0	0.0	0.0
3	52.0	1.0	4.0	142.0	214.0	0.0	2.0	140.0	1.0	2.0
4	59.0	1.0	4.0	140.0	274.0	1.0	2.0	163.0	0.0	1.2
5	63.0	1.0	3.0	140.0	267.0	0.0	2.0	165.0	1.0	0.0
6	62.0	1.0	3.0	130.0	207.0	0.0	0.0	154.0	1.0	3.8
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	1.0	3.0	120.0	245.0	0.0	0.0	158.0	0.0	1.2
2	45.0	1.0	3.0	130.0	335.0	0.0	0.0	152.0	1.0	2.6
3	63.0	1.0	4.0	110.0	206.0	1.0	0.0	164.0	1.0	0.0
4	64.0	1.0	4.0	130.0	273.0	0.0	0.0	157.0	0.0	1.9
5	67.0	0.0	3.0	128.0	233.0	1.0	0.0	168.0	0.0	0.6
6	58.0	1.0	3.0	120.0	197.0	0.0	0.0	149.0	0.0	0.8
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	52.0	1.0	3.0	100.0	255.0	1.0	2.0	142.0	1.0	1.2
2	50.0	1.0	3.0	130.0	217.0	0.0	0.0	170.0	0.0	0.2
3	55.0	1.0	4.0	160.0	254.0	0.0	2.0	145.0	2.0	2.4
4	55.0	1.0	4.0	120.0	277.0	0.0	0.0	145.0	0.0	0.6
5	60.0	1.0	4.0	150.0	218.0	1.0	0.0	163.0	0.0	0.0
6	64.0	1.0	3.0	140.0	274.0	0.0	0.0	160.0	0.0	0.0
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	42.0	0.0	3.0	140.0	233.0	0.0	2.0	142.0	1.0	0.0
2	59.0	1.0	3.0	120.0	246.0	0.0	0.0	105.0	1.0	0.0
3	68.0	0.0	4.0	120.0	271.0	0.0	0.0	161.0	0.0	0.0
4	48.0	1.0	4.0	160.0	249.0	0.0	0.0	172.0	1.0	3.5
5	61.0	1.0	4.0	130.0	325.0	0.0	2.0	146.0	0.0	0.0
6	54.0	1.0	3.0	130.0	212.0	0.0	2.0	151.0	0.0	1.0
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	54.0	1.0	3.0	150.0	270.0	0.0	0.0	159.0	0.0	0.0
2	64.0	1.0	3.0	152.0	288.0	0.0	2.0	160.0	1.0	1.2
3	46.0	1.0	4.0	140.0	206.0	0.0	0.0	170.0	0.0	0.0
4	43.0	0.0	4.0	128.0	270.0	0.0	0.0	146.0	0.0	0.0
5	69.0	1.0	3.0	130.0	233.0	1.0	2.0	132.0	1.0	1.6
6	44.0	1.0	3.0	120.0	283.0	0.0	2.0	147.0	1.0	1.0
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	1.0	3.0	138.0	233.0	0.0	2.0	155.0	0.0	0.6
2	70.0	0.0	3.0	132.0	233.0	0.0	2.0	179.0	0.0	0.0
3	67.0	0.0	4.0	120.0	234.0	0.0	2.0	163.0	0.0	2.2
4	55.0	1.0	4.0	140.0	239.0	0.0	0.0	145.0	0.0	0.9
5	65.0	1.0	3.0	124.0	196.0	0.0	0.0	161.0	0.0	0.8
6	51.0	1.0	3.0	120.0	244.0	0.0	0.0	152.0	1.0	1.4
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	58.0	1.0	3.0	130.0	203.0	0.0	0.0	150.0	0.0	0.0
2	64.0	1.0	3.0	120.0	214.0	0.0	0.0	160.0	1.0	0.0
3	52.0	1.0	4.0	130.0	288.0	0.0	0.0	162.0	0.0	0.0
4	58.0	1.0	4.0	110.0	216.0	1.0	0.0	147.0	0.0	1.0
5	51.0	1.0	3.0	124.0	239.0	0.0	0.0	169.0	0.0	0.0
6	43.0	0.0	3.0	120.0	263.0	0.0	0.0	158.0	0.0	0.0
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	57.0	0.0	3.0	120.0	254.0	0.0	2.0	132.0	0.0	0.0
2	52.0	1.0	3.0	126.0	269.0	0.0	2.0	174.0	1.0	1.0
3	58.0	0.0	4.0	122.0	228.0	0.0	0.0	132.0	0.0	3.6
4	62.0	1.0	4.0	140.0	330.0	1.0	2.0	150.0	1.0	2.6
5	56.0	1.0	4.0	180.0	233.0	0.0	2.0	182.0	1.0	0.2
6	52.0	0.0	3.0	160.0	235.0	0.0	2.0	131.0	0.0	0.0
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	57.0	1.0	3.0	120.0	318.0	0.0	2.0	180.0	0.0	0.4
2	0.0	1.0	3.0	145.0	234.0	0.0	0.0	154.0	0.0	1.8
3	68.0	1.0	4.0	130.0	244.0	0.0	0.0	162.0	0.0	0.8
4	53.0	1.0	4.0	138.0	309.0	0.0	2.0	154.0	0.0	4.5
5	67.0	1.0	3.0	160.0	234.0	1.0	2.0	141.0	0.0	0.5
6	39.0	1.0	3.0	108.0	249.0	0.0	0.0	144.0	0.0	0.0
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6